

Industrial Internship Report on " Forecasting of Smart city traffic patterns"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project is **Forecasting of Smart City Traffic Patterns** is a Machine Learning project that predicts future traffic flow using historical traffic data. It analyzes factors such as time, date, and traffic volume to forecast congestion. The system helps improve traffic management, reduce delays, and support smart city transportation planning.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This six-week internship was a valuable learning experience that helped me improve my technical knowledge and practical skills. Throughout the internship, I worked on the **Forecasting of Smart City Traffic Patterns** project, which focuses on predicting future traffic flow using machine learning techniques and historical traffic data to support smart city transportation management.

This internship highlighted the importance of industry-oriented training in career development by providing hands-on experience with real-world problems and enhancing my problem-solving, programming, data analysis, and machine learning skills.

The opportunity provided by Upskill Campus (USC) in collaboration with UniConverge Technologies (UCT) allowed me to learn through structured weekly tasks, guided learning, and project implementation. The internship was well organized, beginning with the fundamentals of machine learning and data preprocessing, followed by model development, evaluation, and final project submission.

During this internship, I gained knowledge of machine learning algorithms, data preprocessing, feature engineering, model training, performance evaluation, and traffic forecasting techniques. I also improved my documentation, presentation, and project development skills. Overall, this internship increased my confidence and prepared me for future opportunities in machine learning and smart city applications.

I sincerely thank the mentors and team of Upskill Campus (USC) and UniConverge Technologies (UCT) for their continuous guidance and support throughout the internship. I also express my gratitude to my teachers, family, and friends for their encouragement and motivation.

My message to my juniors and peers is to make the best use of internship opportunities, stay dedicated, continue learning emerging technologies, and apply your knowledge through practical projects. Such experiences help develop technical expertise, problem-solving abilities, and confidence for a successful career.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



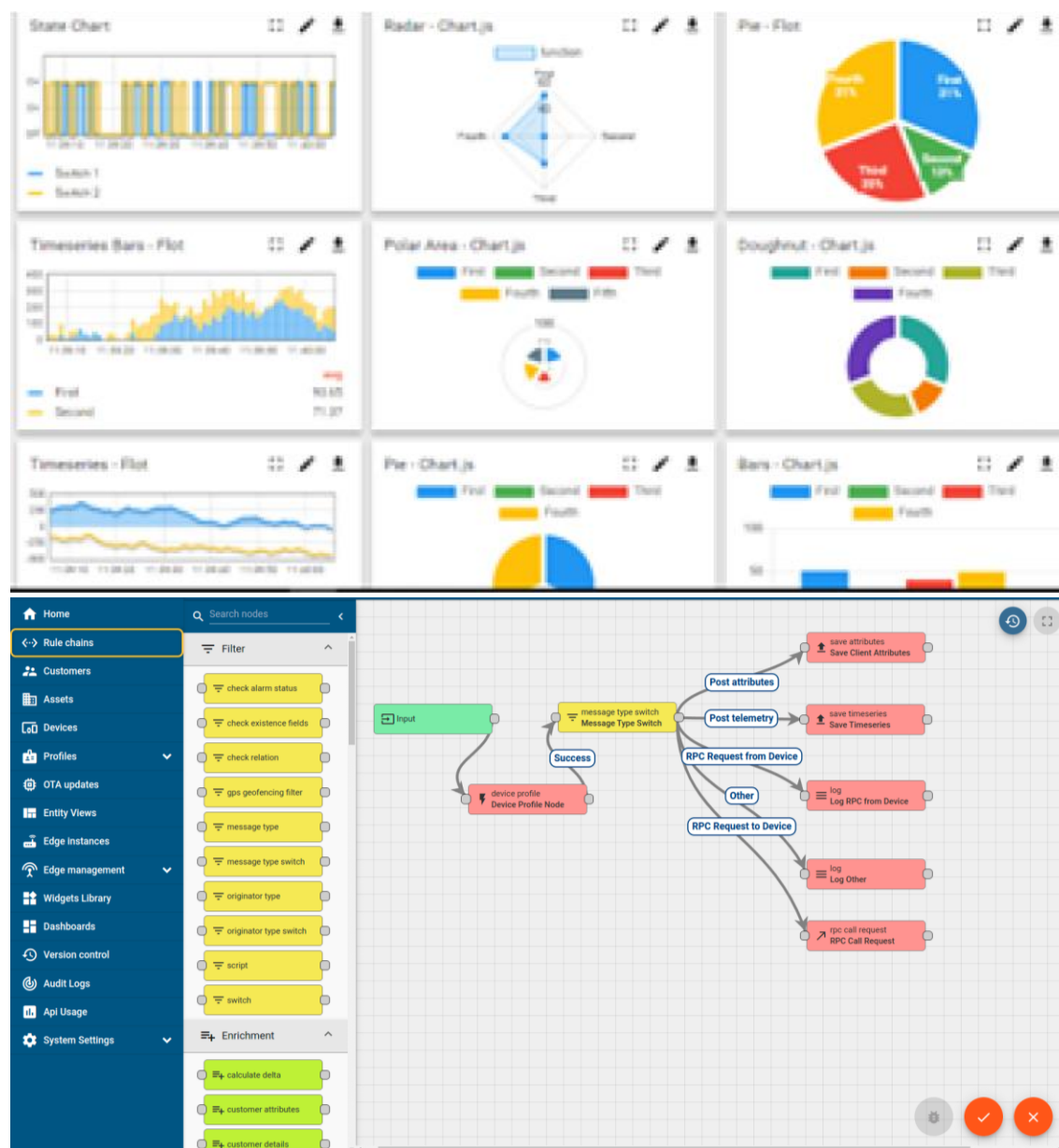
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



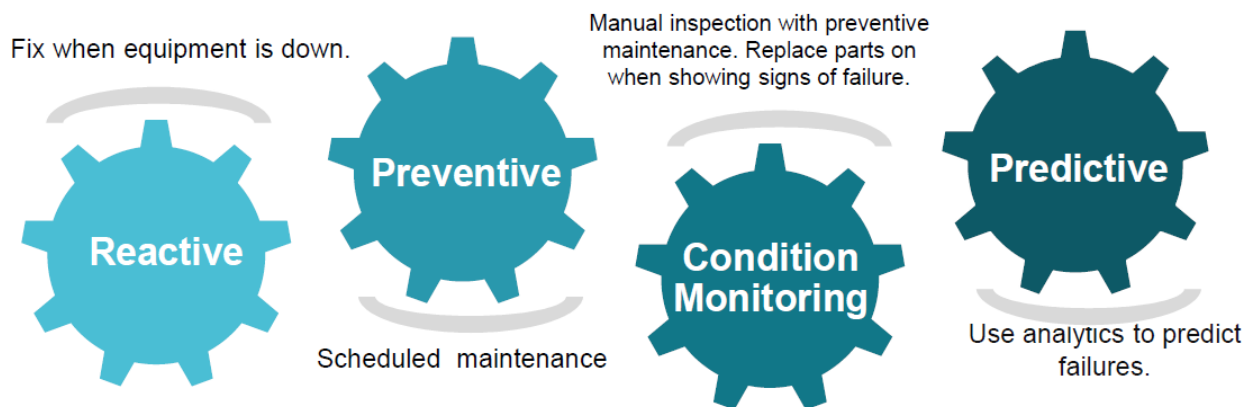


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

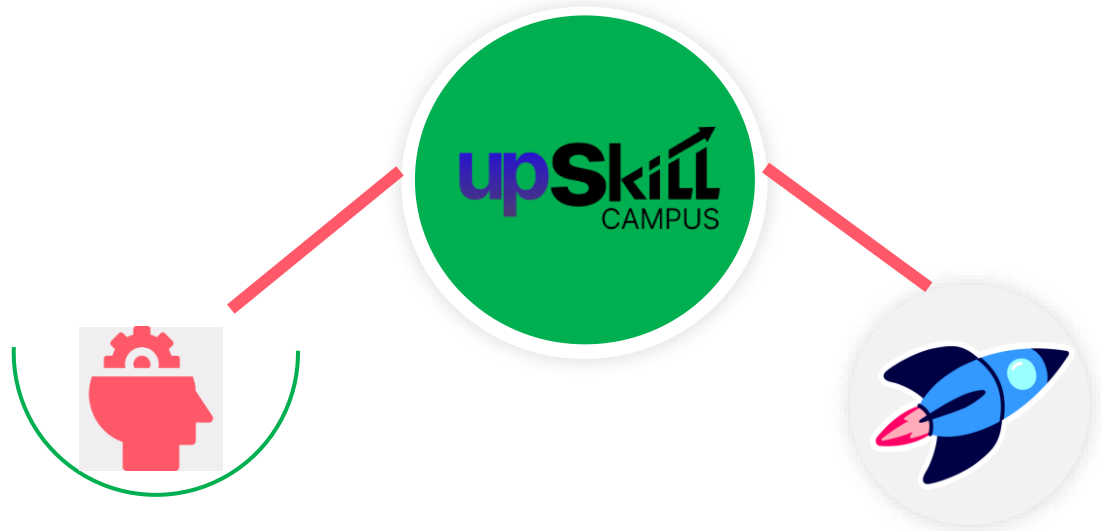
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

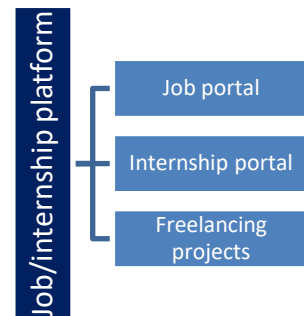
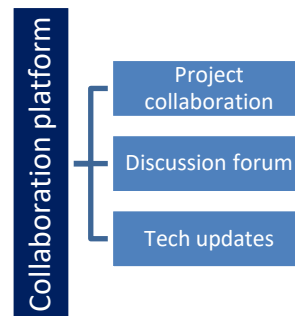
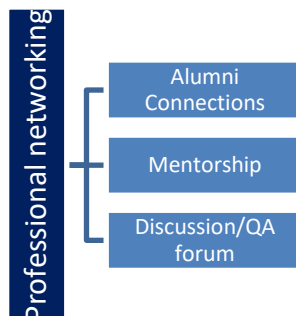
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- To collect and preprocess historical traffic data for accurate analysis.
- To develop a machine learning model for predicting future traffic patterns.
- To analyze the effect of time, date, and other factors on traffic flow.
- To improve traffic management by forecasting congestion in advance.
- To support smart city planning through data-driven traffic predictions.

2.5 Reference

- [1] Aurélien Géron, ***Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow***, 3rd Edition, O'Reilly Media, 2022.
- [2] Christopher M. Bishop, ***Pattern Recognition and Machine Learning***, Springer, 2006.
- [3] Ian Goodfellow, Yoshua Bengio, and Aaron Courville, ***Deep Learning***, MIT Press, 2016.

2.6 Glossary

Terms	Acronym
Machine Learning(ML)	A branch of AI that enables systems to learn from data and make predictions.
Traffic Forecasting	Predicting future traffic flow using historical traffic data.
Dataset	A collection of traffic related data used for training and testing the model.
XGBoost	An optimized gradient boosting algorithm known for high performance and efficiency.
Model Evaluation	Measuring the performance of a model using metrics such as MAE, RMSE and R ² Score.

3 Problem Statement

With the rapid growth of urban populations and vehicles, traffic congestion has become a major challenge in smart cities. Existing traffic management systems often rely on real-time monitoring and are unable to accurately predict future traffic conditions. This project aims to develop a machine learning-based traffic forecasting system that analyzes historical traffic data to predict future traffic flow and congestion levels. The proposed system will help improve traffic management, reduce delays, and support efficient transportation planning in smart cities.

4 Existing and Proposed solution

Existing Solution

Current traffic management systems mainly rely on fixed traffic signal timings, manual monitoring, and basic statistical methods. These systems have limited ability to analyze large amounts of historical traffic data and often fail to predict future traffic conditions accurately. As a result, traffic congestion, delays, and inefficient road utilization remain common problems.

Proposed Solution

The proposed system uses **Machine Learning** techniques to analyze historical traffic data and predict future traffic patterns. It includes data preprocessing, feature engineering, model training, and performance evaluation using algorithms such as **Random Forest**, **Gradient Boosting**, and **XGBoost**. The predicted traffic patterns can help optimize traffic management, reduce congestion, improve travel efficiency, and support smart city transportation planning.

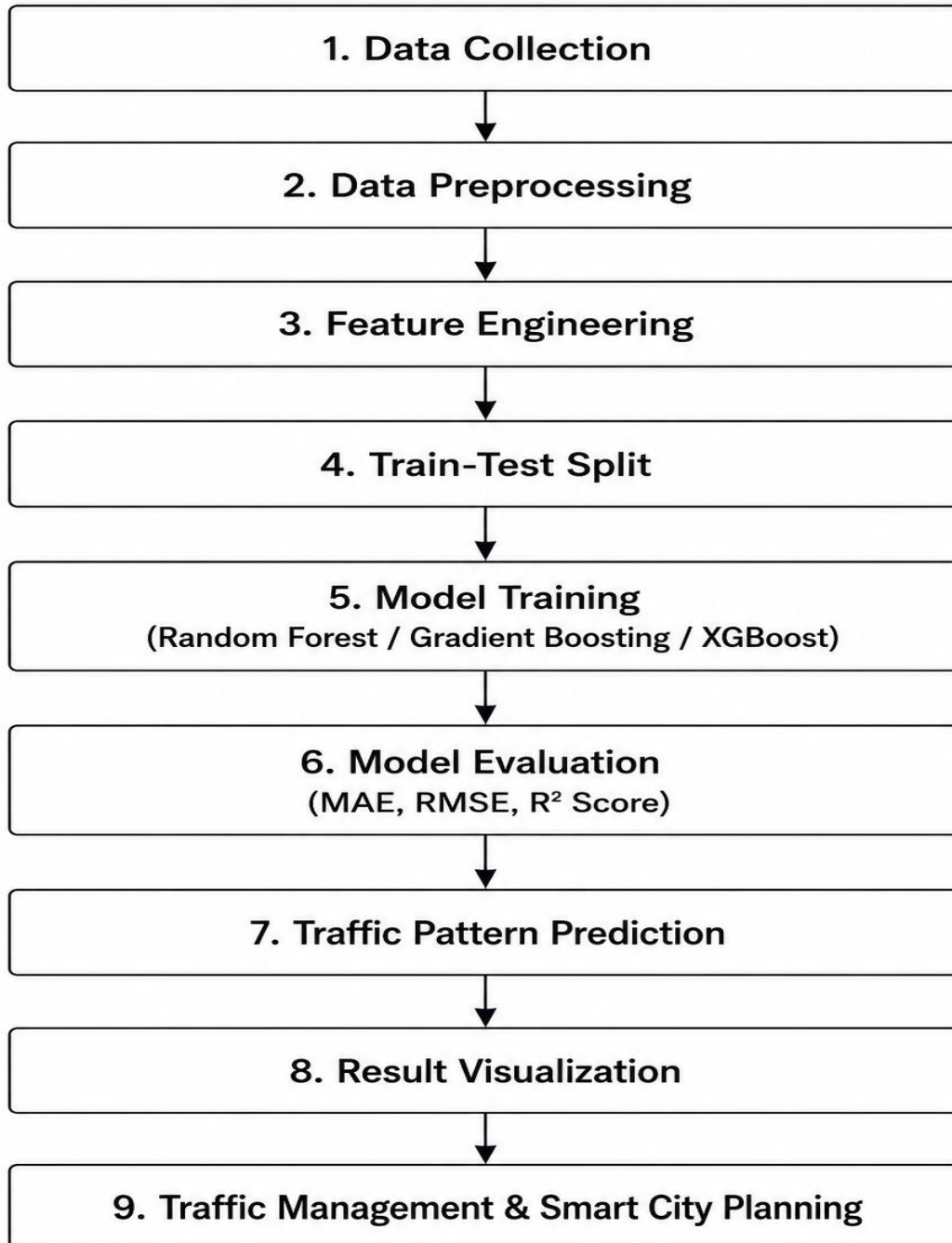
4.1 Code submission (Github link)

<https://github.com/04nik/-Upskillcampus-.git>

4.2 Report submission (Github link) :

https://github.com/04nik/-Upskillcampus-/blob/ce74eac56ca79dbbbfd8ba1f98de94a3b9ec6192/SmartCityTrafficForecasting_Nikhil_USC_UCT.pdf

5 Proposed Design/ Model



6 Performance Test

The performance testing phase evaluates the effectiveness and reliability of the **Forecasting of Smart City Traffic Patterns** system. The objective is to ensure that the proposed machine learning model provides accurate traffic predictions while maintaining efficient processing time for real-world smart city applications.

Several constraints were considered during the development of the system:

- **Prediction Accuracy:** The model should accurately forecast future traffic patterns.
- **Processing Speed:** The system should generate predictions quickly for timely traffic management.
- **Data Quality:** The model's performance depends on the completeness and accuracy of historical traffic data.
- **Memory Usage:** The system should run efficiently on a standard computer without excessive memory consumption.
- **Model Performance:** The selected machine learning algorithm should provide high accuracy with low prediction error.

To address these constraints:

- Data preprocessing techniques such as handling missing values, data cleaning, and normalization were applied.
- Feature engineering was performed to extract meaningful features such as time, date, day of the week, and traffic volume.
- Machine learning algorithms including **Random Forest**, **Gradient Boosting**, and **XGBoost** were trained and compared to identify the best-performing model.
- Model performance was evaluated using standard metrics such as **MAE (Mean Absolute Error)**, **RMSE (Root Mean Square Error)**, and **R² Score**.

The experimental results indicate that the proposed system successfully predicts traffic patterns with high accuracy and low prediction error. The developed model can assist traffic authorities in reducing congestion, improving traffic flow, and supporting efficient smart city transportation planning.

6.1 Test Plan/ Test Cases

Test Case	Objective	Input	Expected Result	Status
TP-01	Verify data pre-processing	Historical traffic dataset	Clean and pre-processed dataset is generated	Passed
TP-02	Verify Future engineering	Preprocessed data	Relevant features are extracted successfully	Passed
TP-03	Verify model training	Training dataset	ML model is trained successfully	Passed
TP-04	Verify traffic prediction	Test dataset	Accurate traffic pattern predictions are generated	Passed
TP-05	Evaluate model performance	Predicted and actual traffic data	Performance metrics (MAE, RMSE, R ² Score) are calculated successfully	Passed
TP-06	Verify final system output	New traffic input data	System predicts future traffic patterns correctly	Passed

6.2 Test Procedure

1. Load the Dataset: Import the historical traffic dataset into the system.
2. Preprocess the Data: Clean the dataset by handling missing values, removing duplicates, and normalizing the data.
3. Feature Engineering: Extract important features such as date, time, day of the week, and traffic volume.
4. Split the Dataset: Divide the dataset into training and testing sets (e.g., 80% training and 20% testing).
5. Train the Model: Train machine learning algorithms such as Random Forest, Gradient Boosting, and XGBoost using the training data.
6. Test the Model: Use the testing dataset to predict future traffic patterns.
7. Evaluate Performance: Measure model performance using MAE (Mean Absolute Error), RMSE (Root Mean Square Error), and R^2 Score.
8. Analyze Results: Compare the predictions with actual traffic data and identify the best-performing model for accurate traffic forecasting.

6.3 Performance Outcome

The proposed **Forecasting of Smart City Traffic Patterns** system successfully predicted future traffic patterns using machine learning algorithms. After training and testing the models, the system achieved good prediction accuracy with low error values. Performance evaluation using **MAE**, **RMSE**, and **R^2 Score** indicated that the selected model produced reliable and consistent results. The system can effectively support traffic management by predicting congestion in advance, optimizing traffic flow, and assisting smart city transportation planning. Overall, the project demonstrated that machine learning is an efficient approach for accurate traffic forecasting and decision-making.

7 My Learnings

During this internship, I gained practical knowledge of **Machine Learning** and its application in traffic forecasting. I learned how to collect, preprocess, and analyze historical traffic data, perform feature engineering, and train machine learning models such as **Random Forest**, **Gradient Boosting**, and **XGBoost**. I also learned to evaluate model performance using metrics like **MAE**, **RMSE**, and **R² Score**.

This internship improved my programming, problem-solving, documentation, and analytical skills. Working on a real-world project increased my confidence and provided valuable hands-on experience that will help me build future machine learning projects and pursue a career in Artificial Intelligence and Data Science.

8 Future work scope

The **Forecasting of Smart City Traffic Patterns** project can be further enhanced by incorporating advanced technologies and additional data sources. In the future, the system can be integrated with real-time traffic data from IoT sensors, GPS devices, and CCTV cameras to improve prediction accuracy. Deep learning models such as LSTM and GRU can be implemented for better time-series forecasting. The system can also include weather conditions, road accidents, public events, and construction activities as additional features for more reliable predictions. Furthermore, the model can be deployed as a web or mobile application to provide real-time traffic updates and route recommendations, supporting intelligent transportation systems and smart city planning.